



PCB Assembly & SMT Line

From bare board to functional product — the modern factory tour

Stencil • Place • Reflow • Inspect • Test

Audience: Engineering Students & Junior Engineers

SMT • THT mixed-process • DFM • Inspection • Final QA



Why PCB Assembly Literacy Matters

Designers who understand the line ship cheaper, better boards

Yield = profit

Each 1% yield drop costs a CM real money.

DFM mindset

Designs become assembly-friendly once you've seen the machines.

Faster bring-up

Spotting defects requires knowing what the process can do.

Vendor dialogue

Talking the CM's language speeds quotes and saves errors.

Cost negotiation

Knowing where time is spent lets you reduce it deliberately.

Quality control

Audit a line to validate before mass production.

Stage 1 — Solder Paste Printing

Stencil + squeegee deposit paste on every pad

- ▶ Steel stencil (laser-cut 0.1-0.15 mm) aligned to PCB
- ▶ Solder paste deposited on top of stencil
- ▶ Squeegee at ~60° forces paste through apertures
- ▶ Stencil lifts → precise paste deposits remain
- ▶ SPI (Solder Paste Inspection) measures every pad
- ▶ Paste height, volume, area, offset checked
- ▶ Defective board returns for re-print

Common print defects

- ▶ Insufficient deposit → open joints after reflow
- ▶ Excess deposit → bridges on fine-pitch ICs
- ▶ Skipped pad → stencil clogged or warped board
- ▶ Smeared paste → squeegee pressure too high
- ▶ Misregistration → paste off-pad → solder balls
- ▶ Paste age → flux activity dropped, poor wetting

Stage 2 — Pick & Place (P&P)

Robotic placement of every SMD component

Feeders

Tape-and-reel, tray, or stick.
Feed components into the head pickup.

Vision system

Cameras verify pickup orientation
and identify part rotation.

Nozzles

Vacuum-pickup, size-matched to part.
Swapped automatically for variety.

Throughput

Hobby: 1-2k CPH (components/hour).
Industrial: 50-100k CPH.

Accuracy

± 0.05 mm placement standard.
 ± 0.025 mm for fine-pitch lines.

Multiple heads

Modern machines have 4-12 nozzles
on a single gantry — parallel pickup.

Stage 3 — Reflow Oven

8-12 zones bake the board through the solder profile

- ▶ Conveyor moves boards through heated zones
- ▶ Each zone independently controlled by IR or convection
- ▶ Profile: preheat → soak → reflow → cool
- ▶ SAC305 peak temp ~245 °C for 60-90 s
- ▶ Sn-Pb peak temp ~210 °C — wider process window
- ▶ Nitrogen-flooded ovens reduce oxidation (better joints)
- ▶ Profile must be characterised for every board design
- ▶ Lid-down sensors record every board's exact thermal history

Profile development

A new board design needs a profile-development run with thermocouples attached. Get it wrong and you get cold joints or component damage at scale.

Lead-free transition

Higher peak temperatures stress components. Spec parts with 260 °C / 10 s capability. ROHS-compliant components are now the global default.

Stage 4 — Automated Optical Inspection (AOI)

Vision-system checks every joint

- ▶ High-resolution camera + ring lights
- ▶ Compares actual board to 'golden' reference
- ▶ Detects: missing parts, wrong polarity, bridges, tombstones
- ▶ Catches 95%+ of visual defects
- ▶ Cannot see under packages (QFN, BGA) — that's X-ray's job
- ▶ False-positive rate depends on programming quality
- ▶ AOI integrates with line traceability (which board failed where)

AOI vs X-ray

- ▶ AOI — visual only, fast (~30 s per board)
- ▶ X-ray — sees hidden BGA balls, slow (~2 min per board)
- ▶ Inline AOI on the conveyor; X-ray usually offline
- ▶ Combination required for BGA-heavy designs
- ▶ Operators verify AOI 'maybe defects' under microscope
- ▶ BGA voids quantified via X-ray (% voiding metric)

Stage 5 — Electrical Test

ICT and FCT verify the board powers up and behaves

ICT — In-Circuit Test

- ▶ Bed-of-nails fixture contacts every test point
- ▶ Measures R, C, V on every node
- ▶ Detects shorts, opens, wrong values
- ▶ Fast (~30 s per board)
- ▶ Requires custom fixture per board (~₹50k-200k)
- ▶ Best for medium-to-high volume

FCT — Functional Test

- ▶ Powers the board, runs firmware-level tests
- ▶ Measures: comms, sensor outputs, LED state
- ▶ Often combined with JTAG / boundary scan
- ▶ Slower (1-5 min per board)
- ▶ Custom firmware + test rig per product
- ▶ Required for safety-critical and shipping QA

Mixed-Process Assembly

When THT components still need to be added

- ▶ Most boards today have a few THT parts (connectors, big caps, transformers)
- ▶ Order: SMT side 1 → SMT side 2 (if double-sided) → THT insertion → wave or hand solder
- ▶ Wave soldering: board floats over molten solder wave — solders all THT pins at once
- ▶ Selective soldering: a small wave moves to each THT joint precisely
- ▶ Hand soldering at the end for ultra-low volume or oddly-shaped parts

Wave soldering

Old but reliable.
Good for high THT pin count.

Selective solder

Robotic mini-wave per joint.
Programmable, flexible.

Hand finishing

Always present for the awkward parts.
Last step before testing.

What the CM Needs From You

Documentation checklist before quoting

Gerber files	RS-274X or X2 — all copper, mask, silk, drill	Pick-and-Place (.pos)	Centroid file — X/Y/rotation/side for every part
BOM (.csv / .xlsx)	Ref-des / MPN / value / package / qty / source	Drill file (.drl)	Excellon drill data — separate from Gerbers
Stencil design	Cream-mask layer if you have custom apertures	Assembly drawing	Visual reference for orientation / fitment notes
Stack-up	Layer count, thickness, materials	Special instructions	Press-fit, mechanical assembly, programming notes

Design for Assembly (DFA) Rules

Layout choices that make assembly cheaper and more reliable

- ▶ Components only on one side if possible (single-sided assembly)
- ▶ Group SMD on top, THT on bottom for mixed-process
- ▶ Keep ≥ 3 mm edge clearance for conveyor handling
- ▶ Fiducials (3 minimum) on every PCB — vision uses them
- ▶ Same orientation for all polarised parts (caps, diodes)
- ▶ Avoid rotational symmetry — pin-1 must be unambiguous
- ▶ Standard component sizes (avoid exotic packages where possible)
- ▶ Test points on top side, ≥ 1.5 mm exposed copper
- ▶ Panelization — group small boards into a 200x250 mm panel
- ▶ Rail/tab break-offs or V-grooves for separation
- ▶ Tooling holes on rails for fixture alignment
- ▶ Avoid hand-solder-only packages on the same side as reflow
- ▶ Spread heat-sensitive parts away from thermal mass
- ▶ Symmetric pad sizes on chip resistors (anti-tombstone)
- ▶ Silkscreen reference designators visible after assembly
- ▶ Avoid copper pour pulling heat from tiny pads

Panelization

Putting multiple boards together for efficient assembly

Why panelize

P&P, reflow, AOI all process panels —
a 50×50 mm board on a 250×300 panel is 30× more efficient.

V-groove

Score line on top and bottom.
Snap apart after assembly.

Mouse-bite tabs

Small drilled tabs hold boards together.
Break off cleanly.

Routed slots

CNC slot defines the board outline.
Tabs hold panel together.

Fiducials

Three per panel + three per board.
Vision uses for alignment.

Border rails

5-10 mm wide rails on top & bottom.
For conveyor + tooling.

Cost Breakdown — What You're Paying For

Where the money goes in a typical SMT order

Stage	Typical % of cost	Driver	How to reduce
Bare PCB	10-20%	Layer count, size, finish	Smaller area; fewer layers; HASL not ENIG
Components	40-60%	BOM value (parts cost)	Source standard parts; design at JLCPCB's preferred list
Assembly setup	10-20%	Programming + feeder setup	Re-orderable; minimise unique part count
Per-board labor	10-15%	Number of parts × placement time	Reduce parts; standardise
Test / inspection	5-10%	ICT fixture + FCT setup	Combine test points; reuse fixture across products
Logistics	5%	Shipping + customs	Order more, ship less often

Prototype rule of thumb

PCB + assembly for a 50×50 mm 4-layer board with 50 unique parts in 5 quantity: ~₹4-8k from JLCPCB.

Common Assembly Defects & Root Causes

What you see on the rework bench

Defect	Symptom	Root cause	Fix
Tombstone	Cap on its end	Asymmetric pad heating	Smaller / symmetric pads
Bridge	Two pins shorted	Excess paste or fine pitch	Smaller stencil aperture
Open joint	No connection	Insufficient paste / cold profile	Better profile / more paste
Missing part	Pad bare	Feeder error or pickup miss	Operator + AOI catch
Wrong polarity	LED, cap reversed	Operator placed in feeder wrong	Mark feeders clearly
BGA void	X-ray shows holes in ball	Paste outgassing during reflow	Different paste; better profile
Solder ball	Loose ball on board	Excess paste squeegee out	Print parameters
Lifted pad	Pad peeled from PCB	Excessive rework heat	Use hot-air not iron; preheat

Domestic vs Offshore Assembly

Where to assemble — trade-offs

India (Bengaluru / Pune / Chennai)

- ▶ Lead time: 1-3 weeks prototypes
- ▶ Cost: ₹4-15k per prototype board (small)
- ▶ MOQ: 1-10 boards possible for prototyping
- ▶ Direct dialogue, site visits possible
- ▶ Customs-free for domestic components
- ▶ Examples: Syrma, Centum, Tessolve, Apollo

China / SE Asia (JLCPCB, etc.)

- ▶ Lead time: 5-12 days for prototype + assembly
- ▶ Cost: \$30-100 / 5 boards (small)
- ▶ MOQ: typically 5+ boards
- ▶ BOM restricted to in-stock parts
- ▶ Customs duty + shipping ~₹2-5k extra to India
- ▶ Examples: JLCPCB, PCBWay, AllPCB, Seeed Fusion

Working with a Contract Manufacturer

How to be a great customer (and get the best service)

- ▶ Send complete documentation (Gerber + BOM + .pos + assembly drawing)
- ▶ Use the CM's preferred parts where possible
- ▶ Specify class (IPC-A-610 Class 1/2/3) and finish
- ▶ Build a relationship — same CM across multiple projects
- ▶ Allow a DFM review round — they catch real issues
- ▶ Approve the 'first article' before mass production
- ▶ Pay on time — small CMs depend on cash flow
- ▶ Test points are mandatory for repeat builds
- ▶ Provide functional test firmware + jig instructions
- ▶ Specify packaging requirements (ESD bag, tray, etc.)
- ▶ Plan for change orders (PCNs, EOLs) ahead of time
- ▶ Get a quote breakdown — line items, not lump sum
- ▶ Buffer 10-15% extra components for losses
- ▶ Document board revisions explicitly (Rev A → Rev B)

Key Takeaways

Key takeaways from this lecture

The line is one machine

Paste → P&P → Reflow → AOI → Test — flows like one process.

Print causes most defects

Optimise paste & stencil first.

DFA = real cost savings

Fiducials, panel rails, symmetric pads pay back every order.

Documentation = quality

Send Gerber + BOM + .pos every time.

CMs need pre-planning

Talk before you ship files — DFM saves weeks.

Visit a line at least once

Even 30 minutes on a CM floor sharpens every future design.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ IPC-A-610 — Acceptability of Electronic Assemblies
- ▶ IPC-7711/21 — Rework / Modification of Electronic Assemblies
- ▶ JLCPCB / PCBWay knowledge bases — practical DFM rules
- ▶ EMSNow.com — industry news on EMS/CM industry
- ▶ SMTA (Surface Mount Technology Association) — courses + papers
- ▶ Phil's Lab (YouTube) — JLCPCB ordering walk-throughs
- ▶ Local CM site visits — ask your professor for an industry tour
- ▶ AS9100 / IATF 16949 — quality systems for aerospace / automotive

Thank you • Build something this week.